AIRSTREET · DESIGN DECISION RECORD

AirStreet concept selection

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/airstreet		
Source	docs/decisions/0004-concept-selection.md @ aa55b15		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	Concept selection among options A, B and C of AST-OPT-001 (kit 1.9.1 upgrade); recommendation A, proposed, awaiting Amish

0004: Concept selection

- **Date:** 2026-10-24
- **Status:** proposed, awaiting Amish. Recommendation: option A, the current design. Recorded in the design decisions register AST-DEC-001 as open decision 1.

Context

Kit 1.8 asks every repo for at least two concept options with different working principles, scored against the user requirements (STANDARDS section 21), and kit 1.9 asks for each to be modelled to the repo's detail level (D3 at P3). AirStreet was designed before kit 1.8, so its current design is option A and is not redesigned without Amish's go. The user requirements are AST-URS-001; the options, their numbers and the scoring are in AST-OPT-001; their science is in AST-STM-001.

Options considered

Option	Working principle	Score (of 260)	Main strength	Main weakness
A, electrochemical and optical node on FieldNode (current)	Naturally ventilated pod with an electrochemical NO2 cell and a laser-scattering PM sensor; passive shield for T and RH; FieldNode solar core; raw signals every 5 min, calibrated by collocation	221	Hourly NO2 and PM2.5 within the pole load, on the lab's shared core	PM2.5 at risk above 85 % RH; field NO2 accuracy unproven (about 4.3 ppb hourly MAE on paper); sensors flagged above 35 degC
B, aspirated, heated-inlet sampling node	Pump draws the sample up a heated inlet past the NO2 cell and the PM sensor; fan-aspirated shield; own 30 W solar supply	188	Dry, steady sample for the PM sensor	About 6 kg and 0.25 m² on the pole; about 28 Wh a day; nitrate loss in the heated inlet
C, passive diffusion tubes with a battery PM logger	Palmes tubes absorb NO2 on triethanolamine for a month and are analysed in a laboratory; battery PM logger read monthly	198	Light, cheap to build, an established method for annual NO2	Monthly NO2 only; a visit at height every month; laboratory cost per tube

C is 10.4 % below A and B is 14.9 % below A, so no option is within 10 % of the leader and a pick is given.

Decision

Proposed, awaiting Amish: carry option A forward as the design of record. `concept_chosen` in `project.yaml` is "A (proposed)" until Amish decides.

Consequences

- If A is confirmed, nothing in the design changes; the drawing, priced BOM and build plan already describe A.

- The runner-up's best idea is proposed separately (AST-DEC-001, open decision 2): triplicate diffusion tubes under a small hood on the cross arm at a few nodes, as an independent check of the annual NO2 means and of the street transfer bias. It adds a hood, a clip and laboratory cost per tube; nothing is changed until Amish decides.
- If Amish prefers C, the repo's pitch changes (no hourly NO2), which is his call; the D3 model of C in `cad/src/ast_options.py` is the starting point.
- Options B and C are modelled to D3 in `cad/src/ast_options.py` and exported as `cad/step/B-assembly.step` and `cad/step/C-assembly.step`.